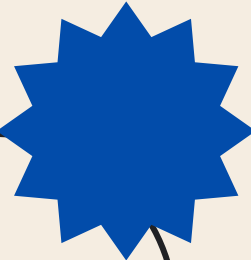




DROP-IN

GROUP 14



DUBAI

Your Gateway to Dubai's Best Experiences

Group 14

Team Members

**Gauri Erunaliyath
Naroth**

**Sandeepkumar
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Durga Anand

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Emlin Bino

INTRODUCTION

- Drop In Dubai is a comprehensive web-based tourism platform designed to simplify travel planning and booking for visitors to Dubai.
- Our application provides an all-in-one solution for discovering and booking hotels, restaurants, attractions, and events across the city.

Key Features

- Browse and book hotels, restaurants, attractions, and events
- User authentication system with role-based access (Admin & Customer)
- Personalised user profiles with favourites and booking management
- Real-time booking management (create, modify, cancel)
- Admin dashboard for content and booking management
- Responsive design for seamless experience across devices

HIGH-LEVEL ■ ARCHITECTURE

- ★ Drop In Dubai follows a three-tier client-server architecture that separates concerns and enables scalability and maintainability.



FRONT END

HTML



JS



CSS



AJAX



Security

BACKEND



Node.js



Express.js

HTTP Requests (JSON via REST API)

JSON Responses (User and Booking Data)

Local Storage



DATABASE



phpMyAdmin

Query Results
(User & Booking Data)

SQL Queries
(Read/Write Operations)

REST
API

Data Exchange



JSON

FRONT END



- HTML for structuring web pages, CSS for styling and layout design, and JavaScript for adding interactivity and dynamic functionality to the user interface.
- Uses Fetch API (AJAX) to send and receive data asynchronously from the backend.
- Displays booking details, listings, and user information dynamically.

Handles UI, user input, and sends requests to the server

Local Storage

- Stores temporary data (like user session info or booking selections) in Local Storage.



Data Exchange

- Data is exchanged between frontend and backend in JSON format.
- The system follows RESTful API principles for structured communication.



Security



- bcrypt secures user passwords through encryption.
- CORS enables safe communication between frontend and backend.

BACKEND



Node.js



Express.js

- Handles all business logic, routing, and API endpoints.
- Processes requests from the frontend and sends responses in JSON format.
- Uses Express.js to manage RESTful routes and middleware.
- Manages booking operations, user registration, and login functionality.
- Connects to the MySQL database using Mysql2 driver.

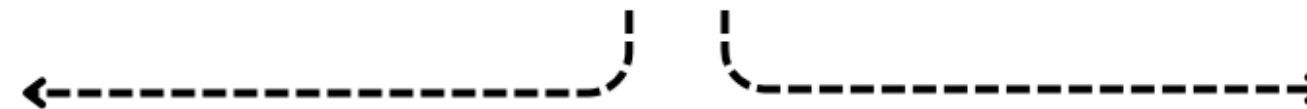
Acts as the bridge between front end and database.

DATABASE

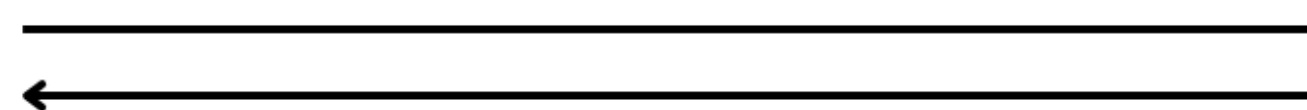


- Stores listings, user info and booking details.
- Managed locally using XAMPP and phpMyAdmin.
- Ensures data consistency and supports CRUD operations (Create, Read, Update, Delete).
- Communicates with the backend through mysql2 driver for data operations

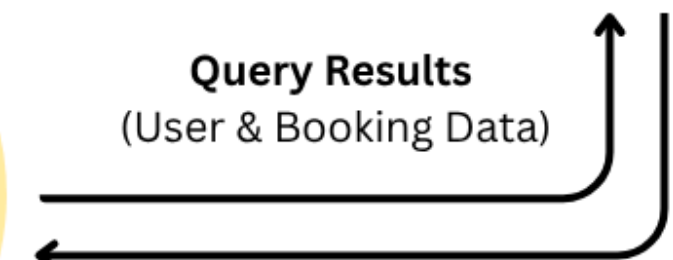
Persistent data storage



HTTP Requests (JSON via REST API)



JSON Responses (User and Booking Data)



Query Results
(User & Booking Data)

SQL Queries
(Read/Write Operations)



User Interface
(Browser)

Request

Response

Internet

Request

Response

Tourist Booking System Application

Web Server

Development Tools

XAMPP

GitHub

VSCode

Backend Layer

Node.js Runtime

Express.js Framework

BCrypt (Password
Encryption)

CORS Middleware

RESTful API Routes

JSON Data Handling

AJAX / Fetch API (JSON Request)

JSON Response
(Booking Info, User Data)

Frontend layer

HTML5

CSS3

Javascript

Fetch API (AJAX)

Local Storage

Query Results
(JSON Data)

SQL Queries
(Read/Write Data)

Databse Layer

- MySQL (RDBMS)
- mysql2 (Node.js driver)
- phpMyAdmin (management)

CODE STRUCTURE

OVERVIEW

1 Frontend Layer

Frontend (Client-Side)
9 HTML pages providing a user interface for all features:

- Authentication (login, signup)
- Main pages (index, profile)
- Booking features (hotels, attractions, events, restaurants)
- Admin dashboard
- Unified CSS styling

2 Application Logic

Application Logic (JavaScript)
12 JavaScript modules handling:

- User authentication & sessions
- Booking management
- Content display & filtering
- Favorites system
- Admin controls
- Dynamic page interactions

Backend & Database 3

Backend & Database
Server infrastructure:

- Node.js Express server
- MySQL database
- RESTful API endpoints
- Image storage
- Session management
- Data validation

CODE STRUCTURE

1

Frontend Layer

Frontend (Client-Side)

9 HTML pages providing a user interface for all features:

- Authentication (login, signup)
- Main pages (index, profile)
- Booking features (hotels, attractions, events, restaurants)
- Admin dashboard
- Unified CSS styling



```
/ (Root Directory)
|—— index.html # Landing page
|—— hotels.html # Hotels listing
|—— restaurants.html # Restaurants listing
|—— attractions.html # Attractions listing
|—— events.html # Events listing
|—— profile.html # User profile & bookings
|—— admin.html # Admin dashboard
|—— login.html # Login page
|—— signup.html # Signup page
|—— style.css # Global styles
|—— /images # Image assets
```

CODE STRUCTURE

2

Application Logic

Application Logic (JavaScript)
12 JavaScript modules
handling:

- User authentication & sessions
- Booking management
- Content display & filtering
- Favorites system
- Admin controls
- Dynamic page interactions

```
/js (JavaScript Modules)
|—— main.js # Navigation & utilities
|—— index.js # Landing page logic
|—— login.js # Login functionality
|—— signup.js # Registration logic
|—— profile.js # Profile management
|—— booking.js # Booking system
|—— favourites.js # Favorites management |——
hotels.js # Hotels page logic
|—— restaurants.js # Restaurants logic
|—— attractions.js # Attractions logic
|—— events.js # Events logic
|—— admin.js # Admin panel controls
```

CODE STRUCTURE

Backend & Database

Backend & Database
Server infrastructure:

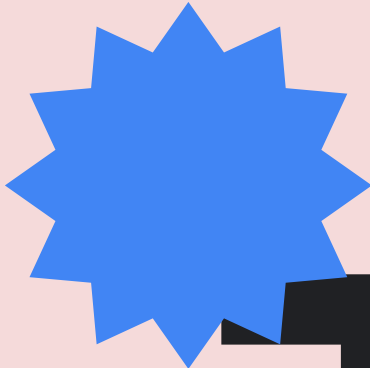
- Node.js Express server
- MySQL database
- RESTful API endpoints
- Image storage
- Session management
- Data validation

3



```
/backend
├── server.js # Node.js Express server
├── package.json # Dependencies
├── package-lock.json # Dependency lock file
└── bookingsystemdb.sql
    # MySQL database schema
```

Database Tables: - users, hotels, restaurants -
attractions, events, bookings - favourites,
admins



Thank You



for listening

"Simplifying Travel - One Booking at a Time"